02/04/26

STOICHIOMETRY

(Measurement of atoms.)

Matter:- Any substance that has mass and occupies space is known as matter.

On the basis of physical properties:-

- **Solids** → Definite shape and fixed volume.
- **Liquids** → No fixed shape. They have fixed volume.
- **Gases** → Neither have a fixed shape nor fixed volume.

Solids, liquids, and gases co-exist at triple point.

On the basis of Chemical composition:-

- **Matter**
  - **Pure** → Same composition
    - **Elemental** → Ex:- He, O_2, N_2
    - **Molecular** → Ex:- Pure water, CO_2, NaCl
      - **Ionic**
      - **Covalent**
  - **Impure** → Mixture
    - **Homogeneous** → Uniform composition → Ex: Salt in Water.
    - **Heterogeneous** → Different phases in the mixture. (Non-uniform)
      - **Colloid** → Ex: Milk, blood.
      - **Suspension** → Ex: Sand in Water.

* Dalton's Atomic Theory:-

⇒ Atoms are the smallest invisible particles present in a substance.

⇒ Atoms of an element have similar size and properties.

⇒ The properties of one element are different from another.

⇒ In a reaction, re-arrangement of atoms takes place. I.e, no atom is created or destroyed in an atom reaction.

* Limitations :-

⇒ Atoms are not the smallest particles in a substance.

⇒ Atoms are made up of electrons, protons, neutrons, positron, etc.

⇒ Electrons, protons, neutrons are the fundamental sub-atomic particles present in an atom.

	Charge			Mass
	Electro-Static unit	Coloumb	relative charge	
Electron	-4.8×10^{-10}	-1.6×10^{-19}	-1	9.1×10^{-31} kg
Proton	4.8×10^{-10}	1.6×10^{-19}	1	1.673×10^{-27} kg
Neutron	0	0	0	1.675×10^{-27} kg

$m_n \cong m_p >>> m_e$

$\frac{m_p}{m_e} = 1836$. i.e, $m_p = 1836\, m_e$.

03/04/26

* Laws of chemical combination :-

(i) Law of conservation of mass :-

In a reaction, total mass of ~~reactions~~ reactants consumed is equal to the total mass of products formed.

(OR)

Mass can neither be created nor destroyed, but it can be transformed.

Ex: 1) $HCl + NaOH \longrightarrow NaCl + H_2O$.

Initial stage: 36.5 gms + 40 gms → 0 + 0.

Final stage: 0 + 0 → 68.5 gms + 18 gms.

Total mass of reactants = 76.5 gms
Total mass of products = 76.5 gms.

Hence, the law is proved.

2) $CaCO_{3(s)} \longrightarrow CaO_{(s)} + CO_{2(g)}$

Initial Stage: 100 gms → 0 + 0.

Stage 2: 20 gms → 44.8 gms + 35.2 gms.

Final stage: 0 → 56 gms + 44 gms.

At initial stage, Total mass of reactants = Mass of product + Mass of unreacted reactant

At final stage. Total mass of reactants = Total mass of prod

* Law of definite proportion / fixed proportion:-

According to the law of defenite proportion, the ratio of mass of different elements combined to form a substance is fixed, irrespective of the method of preperation or the source.

Ex: 1) $H_{2(g)} + 1/2\ O_{2(g)} \longrightarrow H_2O_{(l)}$

1 : 8

$CH_{4(g)} + 2O_{2(g)} \longrightarrow CO_{2(g)} + 2H_2O_{(l)}$

1 : 8

$HCl + NaOH \longrightarrow NaCl + H_2O$

1 : 8

2 parts of hydrogen, i.e, 2 gms of H combines with 1 part of Oxygen, i.e, 16 gms of O.

H : O
2 : 16
1 : 8.

2) $C_{(s)} + O_{2(g)} \longrightarrow CO_{2(g)}$

$CH_4 + 2O_{2(g)} \longrightarrow CO_{2(g)} + 2H_2O_{(l)}$

1 part of carbon, i.e, 12 gms of carbon combines with 2 parts of oxygen, i.e, 32 gms of Oxygen.

CO_2
12 : 32
3 : 8.

* Law of multiple proportions :-

When two substances combine to form two or more different substances, then mass of substances combined will have a simple ratio/ multiples of a simple ratio.

(OR)

When two different elements combine to form two or more substances, then the amount of one of the elements combined fixes the mass of the second element; will be a simple multiple of the first element.

Ex: 1)

$C + O_2$ → CO 12 : 16 = 3 : 4

$C + O_2$ → CO_2 12 : 32 = 3 : 8

Ratio of Oxygen :

4 : 8

1 : 2

2)

$S + O_2$ → SO_2 32 : 32 = 1 : 1 = 2 : 2 = 6 : 6

$S + O_2$ → SO_3 32 : 48 = 2 : 3 = 4 : 6

Ratio of S :

SO_2	SO_3
6	4
3	2

3)

	N	O
N_2O	28	16
N_2O_3	28	48
N_2O_4	28	64
N_2O_5	28	80

Ratio of Oxygen combined :

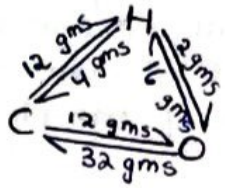
16 : 48 : 64 : 80

1 : 3 : 4 : 5

* Law of reciprocal proportions :-

when two different elements combined with fixed mass of the third element to form two different compounds, then the mass of the elements combined will be either a simple ratio or multiples.

Ex: 1)



$C + 4H \rightarrow CH_4$

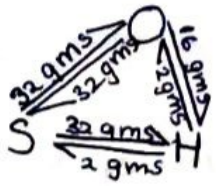
$O + 2H \rightarrow H_2O$.

$C + O_2 \rightarrow CO_2$

3 : 8

12 : 32 { 12 gm C ≙ 4 gm H
3 : 8 { 32 gm O ≙ 4 gm H

2)



$S + O_2 \rightarrow SO_2$

$H_2 + O \rightarrow H_2O$

$S + H_2 \longrightarrow H_2S$

S : H
32 : 2
16 : 1.

S : H
32 : 4
8 : 1 { 32 gm S ≙ 32 gm O
4 gm H ≙ 32 gm O

Q) 1·8 gm of a certain metal burnt in Oxygen gave 3 gm of its oxide. 1·5 gm of the same metal heated in steam gave 2·5 gm of the same oxide. Show that these results illustrate law of defenite proportion.

$M + O \longrightarrow MO$ (where M = Metal)

1·8 gm 3 gm

$\frac{M}{O} = \frac{1·8}{1·2} = \frac{3}{2}$

Mass of O = 3 − 1·8
= 1·2 gm.

$M + H_2O \rightarrow MO + H_2$

1·5 gm 2·5 gm

Mass of O = 2·5 − 1·5
= 1 gm.

$\frac{M}{O} = \frac{1·8}{1·2} = \frac{3}{2}$

∴ It obeys the law of definite proportio (3:2). Hence proved.

06/04/26

* **Mass number (A)** :- Sum of protons and neutrons in an atom.

$A = z + n$

n = number of neutrons.

$n = a - z$.

	Elements	Atomic number (Z)	Mass number (A)
1)	H	1	1
2)	He	2	4
3)	Li	3	6
4)	C	6	12
5)	O	8	16
6)	F	9	18
7)	Na	11	23
8)	Cl	17	35·5
9)	Cu	20	40

* **Complete Symbol** :-

A → Mass number

X → Symbol of element

Z → Atomic ~~no~~ number.

⇒ **Isotopes**:- Elements having same atomic number but different mass number.

Ex:

	$^{1}_{1}H$ Protium	$^{2}_{1}H$ Deuterium	$^{3}_{1}H$ Tritium
Z	1	1	1
A	1	2	3
N	0	1	2

Note :- Isotopes have similar chemical properties but different physical properties.

	H_2O 2:1	D_2O 2:1	T_2O 2:1
Boiling point →	100°C	101.2°C	101.5°C
Density →	1	>1	>1

Isobars: Elements having same mass number but different atomic number

Ex:

	$^{14}_{6}C$	$^{14}_{7}N$
Z	6	7
N	8	7
A	14	14

{No. of protons = No. of electrons}

Note: Isobars have different physical and chemical properties.

⇒

CH_4	NH_3
1:4	1:3
Gas	liquid

⇒ **Atomicity** - Number of atoms present in one molecule.

* **Isotones** :- Elements having same number of neutrons but different atomic number and mass number.

Ex:

	$^{11}_{5}B$	$^{12}_{6}C$
Z ⇒	5	6
A ⇒	11	12
N ⇒	6	6

Isosters :- Two or more different molecules having same atomicity and same number of electrons.

Ex:

	CO_2	N_2O
Atomicity ⇒	3	3
No. of e's ⇒	22	22

Isoelectronic species :- Elements or ions having same no. electrons but different atomic number.

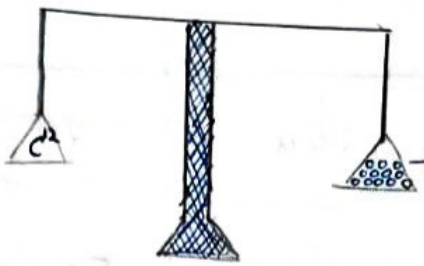
	O^{-2}	F^{-1}	Ne	Na^{+1}	Mg^{+2}
Z ⇒	8	9	10	11	12
e's ⇒	10	10	10	10	10

Atomic mass :- Mass of each atom

Relative atomic mass :- Mass of an atom expressed in terms of ^{12}C is relative atomic mass

⇒ relative atomic mass = $\frac{\text{Mass of each atom}}{\frac{1}{12}(\text{Mass of } {}^{12}C)}$.

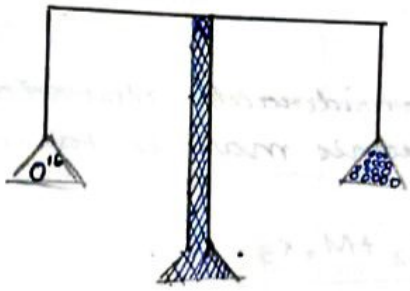
Ex:1)



→ 12 parts of C^{12}.

0 = 1 atomic mass unit (amu)

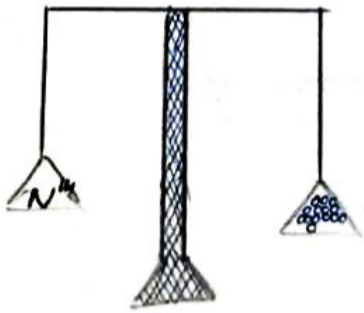
Ex:2)



RAM of O = 16

→ 16 parts of ${}^{12}C$.

Ex:3)



RAM of N = 14

→ 14 parts of ${}^{12}C$.

⇒ In recent days, relative atomic mass (RAM) is being considered as unified mass (u).

⇒ Gram atomic mass : Atomic mass expressed in grams

Element	RAM	Atomic mass	1gram atomic mass.
O	16 amu	16 u	16 gms.
Mg	24 amu	24 u	24 gms
Fe	56 amu	56 u	56 gms

* **Average atomic mass:-**

In case of atoms having considerable abundance of isotopes, then average atomic mass is taken.

$$\text{Avg. Atomic mass} = \frac{M_1x_1 + M_2x_2 + M_3x_3 \cdots}{100}$$

M_1 = Atomic mass of isotopes.
X_1 = Abundance of isotope.

Ex: Cl has two important isotopes of -

^{35}Cl → 75.8 %

^{37}Cl → 24.2 %

$$\text{Avg atomic mass of Cl} = \frac{35(75.8) + 37(24.2)}{100}$$

$$= 35.5$$

07/04/26

* Molecular mass :-

Mass of each molecule, i.e, algebraic sum of atomic mass of each element present on the molecule.

Substance	Molecular mass
O_2	$2 \times 16 = 32$
H_2O	$2(H)+1(O) = 2(1)+1(16) = 18$
CO_2	$1(C)+2(O) = 1(12)+2(16) = 44$
$C_6H_{12}O_6$	$72+12+96 = 180$
H_2SO_4	98
urea - NH_2CONH_2	60
HnO_3	63
C_2H_5OH	46
Mohr salt - $(NH_4)_2SO_4 \cdot FeSO_4 \cdot 6H_2O$	392

* Molar mass :-

Mass of one mole in grams of a substance

-OR-

Algebraic sum of mass number of atoms present in one molecule

	Molar mass	molecular mass	gram molecular mass
H_2O	18 u	18 amu	18 gms
CH_4	16 u	16 amu	16 gms
HNO_3	63 u	63 amu	63 gms

* Mole Concept :- (stoichiometry)

It is the basic SI unit of a substance which contains avagadro number of particles, i.e, 6.023×10^{23}

$$n = \frac{\text{mass of the substance (gms)}}{\text{molar mass}} \longrightarrow (1)$$

(ii) No. of moles in terms of volume :-

At STP (Standard temperature and pressure); i.e, $T = 0^\circ C$, $P = 1$ atm, $T = 273.15 K$.

$T_{mK} = T_C^\circ + 273.15$

$10^\circ C = 283.15 K$

$25^\circ C = 297.15 K$

$-40^\circ C = 233.15 K$

* 1 bar = 1.013 atm
* 1 atm = 760 mm of Hg
  = 76 cm of Hg.
* 1 atm = 760 Torr
* 1 torr = 1 mm of Hg
* 1 bar = 1.013×10^5 dynes/cm^2



* 1 litre = 1000 ml
* 1 litre = $1 dm^3 = 1000 cm^3 = 1000$ ml
* 1 metere = 10 decimetre
* 1 decemetre = 10 cm = 100 mm
* 1 $met^3 = 1000 dm^3 = 1000$ litre

⇒ At STP, i.e, T = 273.15k and P = 1 bar, any gas of one mole occupies 22.7 litres.

⇒ At 273.15 k and 1 atm, any gas of 1 mole occupies 22.4 litres

⇒ 1 mole $\hat{=}$ 22.4 litres

n mole $\hat{=}$ V(lt)

$$\Rightarrow n = \frac{V(lt)}{22.4}$$

⇒ At 1 bar, $n = \frac{V(lt)}{22.7}$

$$\Rightarrow n = \frac{V(ml)}{22400}$$

(iii) **no. of moles in terms of no. of particles** :-

1 mole of any substance contains 6.023×10^{23} number of particles.

Mass of one ^{12}C atom $= 1.992 \times 10^{-23}$ gram

Each part of carbon atom $= \frac{1.992 \times 10^{-23}}{12}$

$= 1.67 \times 10^{-24}$ gms.

$$\frac{1}{1.67 \times 10^{-24}} = 6.023 \times 10^{23}$$

12 gms of ^{12}C contains 6.023×10^{23} number of atoms.





Q) (i) Calculate the no. of molecules present in 16 gm of CH_4

Sol) no. of molecules $= n \times N_A$

$$n_{CH_4} = \frac{16}{16} = 1$$

$$= 1 \times 6.023 \times 10^{23}$$
$$= 6.023 \times 10^{23}$$

(ii) Calculate the volume occupied by 3.01×10^{20} molecules of CO_2 at STP.

Sol) $V_{(lt)} = n \times 22.7$

$$n = \frac{3.01 \times 10^{20}}{6.023 \times 10^{23}}$$

$$= 0.5 \times 10^{-3}$$
$$= 5 \times 10^{-4}$$

$$V(lt) = 5 \times 10^{-4} \times 22.7$$

$$= 113.5 \times 10^{-4}$$

(iii) What is the amount of $CO_{2(g)}$ released by decomposition of 50 gm of $CaCO_3$

Sol) $CaCO_{3(s)} \longrightarrow CaO_{(s)} + CO_{2(g)}$

$$n_{caco_{3(g)}} \cong n_{CO_2}$$

$$n_{caco_3} = \frac{50}{100} = 0.5$$

$$n_{CO_2} = 0.5$$

Mass (gm) $= n$ (molar mass)

$$= 0.5(44) = 22 \text{ gms}$$



(iv) What is the amount of Nitrogen (NH_3) formed in a reaction between 28 gm of Nitrogen and 6 gm of Hydrogen.

Sol) $N_2 + 3H_2 \rightarrow 2NH_3$

1 3 2

$n_{N_2} = \frac{28}{28} = 1$

$n_{H_2} = \frac{6}{2} = 3$

no. of moles of NH_3 formed = 2

Mass of NH_3 = n (molar mass)

= 2(17)

= 34 gms.

~~08/04/26~~

* Limiting reagent :-

The substance present in less quantity than required and controls the amount of product formed in a reaction.

Ex :-

	A	+	B	⟶	AB
Exp ① t=0	1		1		0
	0		0		1
Exp ② t=0	2		1		0
	1		0		1

Amount of A left = given - reacted

= 2 - 1 = 1

2 moles of A requires 2 moles of B but is given only 1 mole of B and hence it is the limiting reagent, A is the excess reagent.

⇒ Excess reagent is left unreacted after the reaction.

Ex:2: $H_2 + I_2 \rightarrow 2HI$

t=0	5	③ LR	0
't'	2	0	6

1 mole of I_2 will form 2 moles of HI hence 3 moles of I_2 will form 6 moles of HI.

Ex:3: $N_2 + 3H_2 \rightarrow 2NH_3$

t=0	2	④ LR	0
't'	0.66	0	2.6

1 mole of N_2 requires 3 moles of H_2. Hence, 2 moles of N_2 requires 6 moles of H_2 but is given in 4 moles. Hence, H_2 is the limiting reagent.

$$\left.\begin{array}{l} 3 \text{ mole } H_2 \cong 2 \text{ mole } NH_3 \\ 4 \text{ Mole } H_2 \cong x \text{ mole } NH_3 \end{array}\right\} \text{cross multiply.}$$

$$x = \frac{4}{3} = 1.33$$

$$1 \text{ mole } N_2 \cong 3 \text{ mole } H_2$$
$$x \text{ mole } N_2 \cong 4 \text{ mole } H_2.$$

* Questions :-

(i) 3 moles of N_2 is allowed to react with 6 moles of H_2 to form NH_3. Find the limiting reagent.

Sol) $N_2 + 3H_2 \longrightarrow 2NH_3$

1 mole of $N_2 \cong$ 3 mole of H_2

3 mole of N_2 requires 9 moles of H_2 but is given 6 moles of H_2.

H_2 is the limiting reagent.

(ii) $CaCO_3 + 2HCl \longrightarrow CaCl_2 + H_2O + CO_2$

0.2 mole of $CaCO_3$ is reacted with 0.5 mole of HCl. Find mass of CO_2 formed.

Sol) $CaCO_3 + 2HCl \longrightarrow CaCl_2 + H_2O + CO_2$

(0.2) L.R	(0.5) L.R	0	0	0
0	0.1	0.2	0.2	0.2

1 mole of $CaCO_3$ forms 1 mole of CO_2, then

0.2 mole of $CaCO_3 \cong$ 0.2 mole of CO_2

Mass of CO_2 = n (molar mass)

$= 0.2(44)$

$= 8.8$ grms.

(iii) Maximum mass of sucrose $C_{12}H_{22}O_{11}$ produced by mixing 84 gms of carbon, 12 gms of hydrogen and 56 lit of O_2 at 1 atm and 273 k is?

Sol) $12C + 11H_2 + 11/2 O_2 \longrightarrow C_{12}H_{22}O_{11}$

7	6	2.5	0

$n_C = \frac{84}{12} = 7$, $n_{H_2} = \frac{12}{2} = 6$

$n_{O_2} = \frac{V(\text{lit})}{22.4}$

$= \frac{56}{22.4} = 2.5$

12 mole of C $\hat{=}$ 11 mole of H_2

7 mole of C $\hat{=}$ x mole of H_2

$x = \frac{77}{12} = 6.4$

H_2 is LR w.r to C.

O_2 is also LR w.r to C.

$x = \frac{7(5.5)}{12}$

$= \frac{38.5}{12}$

$= 3.$

12 mole of C $\hat{=}$ 5.5 mole of O_2

7 mole of C $\hat{=}$ x mole of O_2

11 mole of $H_2 \hat{=}$ 5.5 mole of O_2

6 mole of $H_2 \hat{=}$ x mole of O_2

$x = \frac{6(5.5)}{11} = 3$

O_2 is the whole LR in the above condition.

5.5 mole of $O_2 \hat{=}$ 1 mole of sucrose

2.5 mole of $O_2 \hat{=}$ x mole

$x = \frac{2.5}{5.5} = \frac{5}{11}$, $n_{C_{12}H_{22}O_{11}} = \frac{5}{11}$

mass of sucrose = n (molar mass)

$= \frac{5}{11}(342)$

$= 155.5$ gms

* % Yeild :-

The amount of products formed in an experiment is less than the amount of products expected as per balanced reaction due to some errors.

$$\% \text{ Yeild} = \frac{\text{amount of products formed}}{\text{theoritical amount of product}} \times 100$$

(OR)

$$= \frac{\text{experimental amount}}{\text{theoritical amount}} \times 100$$

09/04/26

* Questions :-

1) No. of atoms in the following samples of substances is the largest in:

(i) 127.0 gm of I_2

$$n_{I_2} = \frac{127}{254} = 0.5$$

no. of I atoms $= 2(0.5)N_A$

$= N_A$

(ii) 71.0 gm of Cl_2

$$n_{Cl_2} = \frac{71}{71} = 1$$

no. of atoms $= 2nN_A$

$= 2N_A$

(iii) $n\text{-mg} = \frac{48}{24} = 2$

$= 2N_A$



(iv) $n_{H_2} = \frac{4}{2} = 2$

$= 2(n_{N_A})$

$= 4N_A$

Q) Which of the following has maximum no. of molecules for the same amount of substance present in each

A) CH_4 (16) B) SO_2 (64) C) N_2 (28) D) O_2 (32)

no. of molecules $= n \cdot N_A$

$n = \frac{\text{mass (gms)}}{\text{molar mass}}$

no. of molecules $= \frac{\text{mass (gms)}}{\text{molar mass}} \times N_A$

Q) 5 moles of AB_2 weigh 125×10^{-3} kg and 10 moles of A_2B_2 weighs 300×10^{-3} kg. The atomic mass of A and B in kg mol^{-1} is:

1) $M_A = 50 \times 10^{-3}$, $M_B = 25 \times 10^{-3}$

2) 25×10^{-3}, 50×10^{-3}

3) 5×10^{-3}, 10×10^{-3}

4) 10×10^{-3}, 5×10^{-3}

Sol) 5 moles of $AB_2 = 125 \times 10^{-3}$ kg

1 mole of $AB_2 = \frac{125 \times 10^{-3}}{5} = 25 \times 10^{-3}$

$A + 2B = 25 \times 10^{-3}$ ——— (1)

10 moles of $A_2B_2 = 300 \times 10^{-3}$

iv) $n_{H_2} = \frac{4}{2} = 2$

$= 2(n_{N_A})$

$= 4N_A$

Q) Which of the following has maximum no. of molecules for the same amount of substance present in each

A) CH_4 (16) B) SO_2 (64) C) N_2 (28) D) O_2 (32)

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$n = \frac{\text{mass (gms)}}{\text{molar mass}}$

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$A + 2B = 25 \times 10^{-3}$ ——— (1)

10 moles of $A_2B_2 = 300 \times 10^{-3}$

Q) From 2 Mg Calcium, 1.2×10^{19} atoms are removed, the number of gm atoms of calcium left is :

Sol) $n_{Ca} = \frac{2 \times 10^{-3}}{40} = 5 \times 10^{-5}$

no. of Ca atoms $= nN_A = 5 \times 10^{-5} \times 6.023 \times 10^{23}$

$= 30 \times 10^{18} = 3 \times 10^{19}$

Ca atoms left = Given - removed

$= 3 \times 10^{19} - 1.2 \times 10^{19}$

$= 1.8 \times 10^{19}$

no. of gm atoms (OR)

no. of moles $= \frac{\text{no. of particles}}{N_A}$

$= \frac{1.8 \times 10^{19}}{6 \times 10^{23}}$

$= 3 \times 10^{-5}$

Q) A certain compound has the molecular formula X_4O_6. If 10 gms of X_4O_6 has 5.72 gms of X, atomic mass of X is :

In 1 mole of X_4O_6 contains 4 moles of X and 6 moles of O

$n_X : n_O = 4 : 6$

$\frac{5.72}{a} : \frac{4.28}{16} = 4 : 6$

$\left(\frac{5.72}{a}\right) \times 6 = \left(\frac{4.28}{16}\right) \times 4 \quad a = 32.07 \text{ u}$.

34.32 / 1.7

17
34
1

Q) The charge on 1gm-ion of Al^{+3} in:

A) $\frac{1}{27}$ N_A e Coloumbs

B) $\frac{1}{3}$ N_A e Coloumbs

C) $\frac{1}{9}$ N_A e Coloumbs

D) 3 N_A e Coloumbs

* Charge present on each electron $= -1.6 \times 10^{-19}$ C.
Charge present on each proton $= 1.6 \times 10^{-19}$ C.
Total charge on one mole of electron $= 1.6 \times 10^{-19} \times N_A$ C.
$= 96,500$ C

96,500 C $\hat{=}$ 1 Faraday.

10/04/26

Q) 5 moles of $CaCO_3$ on heating yields 2 moles of CO_2. find % yield of reaction

$$\underset{5}{CaCO_3} \xrightarrow{\Delta} \underset{0}{CaO} + \underset{0}{CO_2}$$

$n_{CaCO_3} \hat{=} n_{CO_2}$

$\% \text{ yield} = \frac{2}{5} \times \frac{20}{100}$

$= 40\%$

Q) 245 gm of $KClO_3$ on heating yields 64 gm of O_2 (K = 39, Cl = 35.5)

$2KClO_3 \rightarrow 2KCl + 3O_2$

2 mole of $KClO_3 \hat{=}$ 3 mole of O_2

$n_{KClO_3} = \frac{245}{122.5} = 2$

From 2 moles of $KClO_3$ given 3 mole of O_2 formed.

mass of O_2 = n (molar mass)

= 3(32)

= 96 gms

% yield = $\frac{64}{96} \times 100$

= 66.67 %

Q) A chalk sample requires 17.52 gm of HCl for complete reaction with all $CaCO_3$ present in it. If the chalk sample is 72% pure, the mass of the sample taken is?

% purity = $\frac{\text{effective mass of substance}}{\text{mass of substance given}} \times 100$

$CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$

1 mole of $CaCO_3 \cong$ 2 mole of HCl.

n $\cong \frac{1}{2}$ mole of HCl

$n_{HCl} = \frac{17.52}{36.5} = 0.5$

$n_{(2)} = \frac{1}{2}$

$n_{CaCO_3} = \frac{1}{4}$

$n = \frac{\text{mass}}{\text{molar mass}}$

n (molar mass) = mass

$\frac{1}{4} \times 100$ = mass

25 gms = mass of $CaCO_3$

$$\% \text{ purity} = \frac{\text{effect mass}}{\text{mass of substance}} \times 100$$

$$\frac{72 \times 25}{100} = \text{effect mass}$$

18 gms = effet. mass of $CaCO_3$.

Illustration 28 (pg 25):-

$$Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4 + 2NaCl$$

Sodium sulphate - 3.0 g
Barium sulphate - 1.74 g

Na_2SO_4 + $BaCl_2$ → $BaSO_4$ + 2NaCl
142 g 233 g

1.74 g of $BaSO_4$ = $\frac{71}{233} \times 1.74 = 0.87$

$\frac{116.5}{58.25}$

$$\% \text{ purity} = \frac{1.06}{3.0} \times 100$$

$$= 35.33\%$$

Beginner leax-5 (pg-26):-(Q-8):-

% purity = 80 %

$$CaCO_3 + 2HCl \longrightarrow CO_2 + H_2O + CaCl_2$$
100 g 44 g

1 mole of $CaCO_3$ ≡ 1 mole of CO_2

25 gms of impure $CaCO_3$

effective mass of $CaCO_3$ = 25 (0.8)
= 20 gms

$n_{CaCO_3} = \frac{20}{100} = 0.2$

$n_{CO_2} = 0.2$

At NTP, 273.15 k and 1 atm pressure, a gas occupies 22.4 lt of 1 mole.

$V_{CO_2} = 0.2\ (22.4)$

$= 4.48$ litres

Beginner's box -5 (Q- 9) pg-26:-

$Ag_2S \longrightarrow 2Ag + S$

1 mole of $Ag_2S \cong$ 2 moles of Ag

248 gm $\Rightarrow$ 216 gm

x gm ~~$\Rightarrow$~~ 1gm

$x = \frac{248}{216}$

$y = \frac{248}{216} \times 100 \times 1.34$

1.34 % of $Ag_2S \Rightarrow \frac{248}{216}$ gms

100 % of ore $\Rightarrow$ x gm

$= 85.68$ g



Q) How many hydrogen atoms are present in 5g of ammonia?

no. of atoms $= 3\left(\frac{5}{17}\right) N_A$

$= \frac{15}{17} N_A$

Q) Illustration - 7 - pg - 15 :-

$Na_2CO_3 \rightarrow$ no. of gram molecules $= 53$

$Na = 23 \times 2 = 46$

$C = 12$

$O = 48$

$= 106$

no. of gm molecules $= \frac{53}{106} \frac{1}{2} = 0.5$ grams

no. of molecules $= 0.5 \times N_A$

$= 0.5 \times 6.023 \times 10^{23}$

$= 3.0115 \times 10^{23}$

$= 3.011 \times 10^{23}$

no. of oxygen atoms $= 3$

$= 3 \times 3.011 \times 10^{23}$

$= 9.033 \times 10^{23}$

Q) Illustration - 8 - pg - 15 :-

Mass removed $= \frac{11.2}{22.4} \times 32 + \frac{N_A}{N_A} \times 16$

$= 16 + 16$

$= 32$ g.

* Law of gaseous volume :- (Gussac's law)

In a gaseous reaction, the volume of gases combine to form product will be in a single ratio of volume.

$$H_{2(g)} + Cl_{2(g)} \rightarrow 2HCl_{(g)}$$
1 lt 1 lt 2 lt

applicable at constant T & P.

* Avagadro's law :-

At constant T and P, equal volume of different gases contain equal number of molecules.

H_2 N_2
2 lt 2 lt

∴ molecules of H_2 = no. of molecules of N_2.

$V \propto n$.

15/04/26

* Density:-

For solids and liquids

Absolute density = $\frac{\text{Mass}}{\text{Volume}}$

Relative density = $\frac{\text{Density of Substance}}{\text{Density of pure water at } 4^\circ}$

Density of pure water at $4^\circ = 1$

i.e, 1 litere of water contains 1 kilo-gram of mass of water.

⇒ For Gases :-

Absolute density = $\frac{\text{Mass}}{\text{volume}}$.

Vapour density of gas = $\frac{\text{Molar mass of gas}}{\text{Molar mass of } H_2}$ in terms of H_2

Vap. density of gas = $\frac{\text{Molar mass of } H_2 \text{ Gas}}{2}$.

Molar mass = 2 (Vap. density)

⇒ For solid :- (From Dulong-petit's law)

Product of molar mass and specific heat is approximately equal to 6.4

Molar mass (specific heat) $\cong 6.4$

Molar mass = $\frac{6.4}{\text{Specific heat}}$

* Specific heat :-

The amount of heat energy required to increase the temperature of 1gm substance by $1^\circ C$.

* Empirical formula:-

The smallest ratio of different elements present in a compound

Ex:- $C_6H_{12}O_6$

6 : 12 : 6

1 : 2 : 1

$= CH_2O$

Borazine $\Rightarrow B_3N_3H_6$

3 : 3 : 6

1 : 1 : 2

$\Rightarrow BNH_2$

% Composition of each element in a compound:-

$$\%\text{ of each element} = \frac{\text{no. of atoms (atomic mass)}}{\text{molecular mass}} \times 100$$

Ex:1) H_2SO_4

$$\%\text{ of H} = \frac{2(1)}{98} \times 100 = \frac{2}{98} \times 100$$

$$\%\text{ of S} = \frac{1(32)}{98} \times 100 = \frac{32}{98} \times 100$$

$$\%\text{ of O} = \frac{4(16)}{98} \times 100 = \frac{64}{98} \times 100$$

(OR)

$$\%\text{ of O} = 100\% - \%\text{ of H} - \%\text{ of S}.$$

Ex:2) H_2O

$$\%\text{ of O} = \frac{1(16)}{18} \times 100$$

$$\%\text{ of H} = \frac{2(1)}{18} \times 100$$

(OR)

$$\%\text{ of H} = 100 - \%\text{ of O}$$

* Systematic procedure to deduce emperical formula from % composition of each element.

Step -① :- Consider the given % of each element as mass of the element present in 100 gm of the substance.

Step -② :- Convert the mass of each element to moles.

$$\text{No. of moles} = \frac{\text{Mass (gms)}}{\text{Atomic mass}}.$$

Step -③ :- The simplest ratio of moles of each element is the E.F.

Q) A given compound contains 40% of C, 6.66% of H, and 53.33% of O, then find its E.F.

Step -① :- C = 40 gms H = 6.66 gms O = 53.33 gms

Step -② :- $n_C = \frac{40}{12} = 3.33$ $n_H = \frac{6.66}{1} = 6.66$ $n_O = \frac{53.33}{16} = 3.33$

Step -③ :- C : H : O

3.33 : 6.66 : 3.33

1 : 2 : 1

$= CH_2O$.

* Concentration terms :-

It is the way to express concentration of a homogeneous binary solution

Solute + Solvent → Solution

Solute → Minor
Solvent → Major

Solute :- The substance that gets dissolved in another substance.

Solvent :- The substance that dissolves other substances.

Solution :- The mixture of solute and solvent.

Solubility :- (Concentration)

The maximum amount of substance dissolved in a given solution at constant temperature.

Concentration

Daily life	Laboratory
Mass percentage ($^{w}/_{w}$ %)	Molarity (M)
Volume percentage ($^{v}/_{v}$ %)	Normality (N)
Mass-volume ($^{w}/_{v}$)	Morality (m)
Strength.	Mole fraction (x)

$$\text{Mass percentage } (^{w}/_{w}\%) = \frac{\text{Weigth of solute (gm)}}{\text{weight of solution (gms)}} \times 100$$

$$\text{Volume percentage } (^{v}/_{v}\%) = \frac{\text{Volume of solute (lt)}}{\text{Volume of solution (lt)}} \times 100$$

Mass-volume (W/V %) = $\frac{\text{Weight of solute (gm)}}{\text{Volume of solution (ml)}} \times 100$

Strenght = $\frac{\text{Weight of solute (gm)}}{\text{Volume of solution (lt)}}$

Q) A solution is prepared by adding 5 gm of NaCl to make 100 gm of solution then ~~find~~ weight % of solution:-

Sol)

$$W/W\ \% = \frac{5}{100} \times 100 = 5\%$$

Q) What is the amount of water required to prepared 13 W/W % of solution which contains 10 gms of NaOH.

Sol)

$$W/W\ \% = \frac{W_{solute}}{W_{solution}} \times 100$$

$$13 = \frac{10}{W_{solution}} \times 100$$

$$W_{solution} = \frac{10}{13} \times 100$$

$$= 72.3 \text{ gms}$$

$$W_{solvent(water)} = W_{solution} - W_{solute}$$

$$= 72.3 - 10$$

$$= 62.3 \text{ gms.}$$

* Laboratory concentration terms:-

Molarity (M):- Number of moles of solute present per litre of solution.

$$M = \frac{n\,solute}{n\,Solution\,(l)} \rightarrow (1)$$

$$M = \frac{n\,Solute \times 1000}{V\,Solution\,(ml)}$$

$$M = \frac{Weight\ of\ Solute\ (g)}{Molar\ mass\ of\ solute} \times \frac{1000}{V\,Solution\,(ml)}$$

Q) What is the molarity of 2mole $CaCl_2$ in 2lt Solution?

Sol) $M = \frac{2}{2} = 1$ Mole lt.

Q) What is the molarity of solution prepared by adding 4.9 gm of H_2SO_4 in 500ml solution

Sol) $W_{solute} = 4.9$ gm

Molar mass of $H_2SO_4 = 98$

$$M = \frac{4.9}{98} \times \frac{1000}{500} = 0.1M$$

$$\text{Weight of Solute (gm)} = \frac{M(\text{Mol. mass})(\text{Solute}) \times V_{sol}(ml)}{1000}$$

Q) Mass of $KMnO_4$ required to prepare 0.02 M Solution of 250 ML (given mol. mass of $KMnO_4$ = 158)

Sol) $$W_{Solute} = \frac{(0.02)\,158 \times 250}{1000} = \frac{1.58}{2} = 0.79\ gms.$$

* Principle of Dilution :-

During dilution, no. of moles of solute remains same before and after dilusion.

$$M = \frac{n}{V(lt)} \Rightarrow n = M \cdot V_{(lt)}$$

$$n_{before} = n_{After}$$

$$M_1V_1 = M_2V_2$$

Q) What is the amount of water to be added to prepare 0.03 M HCl of 100 ml from 0.1 M HCl.

Sol) Before dil^n

$M_1 = 0.1M$

$V_1 = ?$

After dil^n

$M_2 = 0.03$

$V_2 = 100ml$

$M_1V_1 = M_2V_2$

$$V_1 = \frac{M_2V_2}{M_1} = \frac{(0.03)(100)}{0.1}$$

$= 30ML.$

Amount of water added = $V_{final} - V_{initial}$.

$= 100 - 30$

$= 70ml.$

To the 0.1 M HCl of 30ml, if 10 ml of water is added, then it will make up 0.03 M solutions of 100 ML.

* Relation between M & M/v% :-

$$W/v\% = \frac{W_{solute}\ (gms)}{Volume\ of\ solution\ (ml)} \times 100$$

$$M = \frac{W_{solute}}{m \cdot M_{solute}} \times \frac{1000}{V_{solution\ (ml)}}$$

$$M = \frac{(W/v\%)\,10}{m \times m_{solute}}$$

* Relation between M & W/W% :-

$$W/W :- \frac{W\ Solute\ (gm)}{W_{Solution}\ (gms)} \times 100$$

$$Density\ of\ solution = \frac{W_{solution}\ (gm)}{V_{solution}\ (ml)}$$

$$V_{solution\ (ml)} = \frac{W_{Solution}\ (gm)}{D\ (gm/ml)}$$

$$M = \frac{W_{solute}}{M \cdot M_{solute}} \times \frac{1000}{V_{solution\ (ml)}}$$

$$= \frac{W_{solute} \times 1000}{m \cdot M_{solute} \cdot W_{solution}} \; D\ (gm/ml)$$

$$M = \frac{(W/w\%) \cdot D\ gm/ml \times 10}{m \times M_{solute}}$$

Pg-46- Q) 5):-

$$W_{solute} = \frac{M \times (M.M) \times V_{(ml)}}{1000}$$

$M.M = 85$

$V = 50$

70 mg of Na $\longrightarrow$ 1ml

Mass of Na $= 70(50)$

$= 3500$ mg.

Weigth of Na $= 3.5$ gm

23 gm of Na $\rightarrow$ 1 mole of $NaNO_3$

3.5 gm of Na $\rightarrow$ x moles.

$$x = \frac{3.5}{23} = 0.15$$

$n_{NaNO_3} = 0.15$

$$n = \frac{Mass}{mol.mass}$$

Mass $= n$ (mol. mass)

$= 0.15 \times 85$

$= 12.75$

$= 13$ gms.

Beginner box-4

3) i) $NH_3 + O_2 \rightarrow NO + H_2O$

mole) $\Rightarrow 2NH_3 + 5/2 O_2 \rightarrow 2NO + 3H_2O$

2 mole of NH_3 reacts with 2.5 mole O_2

0.4 mole $NH_3 \cong$ x mole O_2

$$x = \frac{2.5(0.4)}{2} = \frac{2.5(0.2)}{= 0.5}$$

$$= 0.5$$

Q3) ~~$I_2 + 2Cl_2 \rightarrow ICl + ICl_3$~~

Sol) $2P_4 + 8O_2 \rightarrow P_4O_6 + P_4O_{10}$

$$n_{P_4} = \frac{31}{4(31)} = 0.25$$

$$n_{O_2} = \frac{32}{32} = 1$$

0.25 mole of P_4 contains 1 mole of P atom.

a moles of P in P_4O_6

b moles of P in P_4O_{10}

$$4a + 4b = 1 \rightarrow (1)$$

$$n_{I_2} = \frac{\text{Mass}}{\text{Mol. mass}} = \frac{25.4}{254} = 0.1$$

$$n_{Cl_2} = \frac{14.2}{71} = 0.2$$

1 mole of O_2 contains 2 moles of O atom

$$6a + 10b = 2 \rightarrow (2).$$

$6 \times (1) - 4 \times (2)$.

$$24a + 24b = 6$$

$$24a + 40b = 8$$

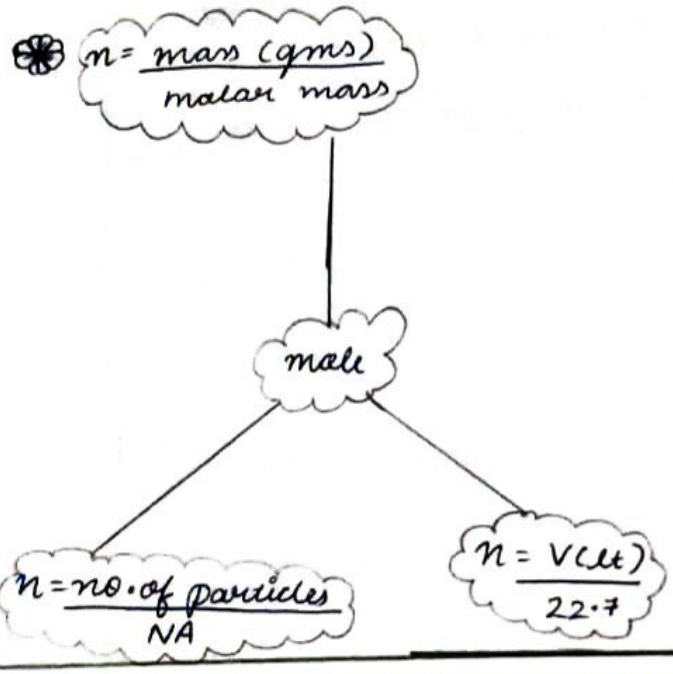
$$16b = 2$$

$$b = \frac{1}{8}$$

⇒ 1 mole $\cong$ 6.023×10^{23} (N_A)

n mole $\cong$ no. of particles

$$n = \frac{\text{no. of particles}}{6.023 \times 10^{23}}$$



Q) Calculate the no. of moles present in the following:-

(i) 16 gms of CH_4 = $n = \frac{\text{mass (gms)}}{\text{molar mass}} = \frac{16}{16} = 1$

(ii) 2 kg of H_2 = $n = \frac{2000}{2} = 1000$

(iii) 5.6 lt of CO_2 at STP of 1 atm = $n = \frac{V(lt)}{22.4} = \frac{5.6}{22.4} = \frac{1}{4} = 0.25$

(iv) 11350 ml of SO_3 at STP = $n = \frac{11.35}{22.7} = 0.5$

(v) 3.01×10^{23} molecules of HNO_3 = $n = \frac{\text{no. of particles}}{N_A}$

$= \frac{3.01 \times 10^{23}}{6.023 \times 10^{23}} = 0.5$ (approx)



Mass of $P_4O_6 = \frac{1}{8} \times 220$

$= 27.5$ gm

Mass of $P_4O_{10} = \frac{1}{8} \times 284$

$= 35.5$ gm.

(1) $\Rightarrow 4a + 4 \times \frac{1}{8} \times \frac{1}{2} = 1$

$4a = \frac{1}{2}$

$a = \frac{1}{8}$.

* Molarity is dependent on temperature. With increase in 'T', volume of solution increases and molarity decreases.

$M \propto \frac{1}{T}$

* Molarity in case of Mixing of solutions:-

In case of mixing solutions of different concentration,

$$M = \frac{M_1V_1 + M_2V_2}{V_1 + V_2}$$

Q) 0.1M of HCl of 100 ml is mixed with 0.02 M HCl of 300 ml. The resultant concentration is,

Sol) $M = \frac{(0.1)(100) + (0.02)(300)}{400}$

$= \frac{10+6}{400} = \frac{16}{400} = 0.04M.$

* Mixing of solution of opposite nature :-

In case of mixing of reacting substance such as acid and base, one will neutralise the other.

$$M = \frac{M_1V_1 - M_2V_2}{V_1 + V_2}, \text{ where } M_1V_1 > M_2V_2$$

Q) 0.01 M of HCl of 10 ml is mixed with 0.02 M of NaOH of 20 ml. what is the nature of the resultant concentration.

Sol) $HCl + NaOH \longrightarrow NaCl + H_2O$.

$$M = \frac{0.4 - 0.1}{30}$$

$$= \frac{0.3}{30} = 0.01.$$

$$n_{HCl} = \frac{M \cdot V(ml)}{1000}$$

$$= \frac{0.01 \times 10}{1000}$$

$$= 10^{-4}$$

$$n_{NaOH} = \frac{(0.02)(20)}{1000} = 4 \times 10^{-3}$$

---

* Normality :- (N) (will be studied further during redox)

Number of equivalents of solute present per litre of solution.

$$N = \frac{\text{no. of equivalents of solute}}{\text{volume of solution (lt)}}$$

Units :- $Eq \cdot lt^{-1}$

* Molality (m) :-

Number of moles of solute present per kg of solvent.

$$n = \frac{\text{no. of moles of solute}}{\text{weight of solvent (kg)}}$$

Units :- $Mole \cdot kg^{-1}$

$$m = \frac{\text{Weight of solute (gms)}}{\text{molar mass of solute}} \times \frac{1000}{\text{weight of solvent (gms)}}$$

$$W_{solute} = \frac{m(M \cdot M)\, W_{solvent}(gms)}{1000}$$

* Relation between M and m :-

Density of solution $(gm/cm^3) = \frac{\text{Mass of solution (gms)}}{\text{Volume of solution (ml)}}$

M = Molarity, m = Molality

$$m = \frac{n_{solute} \times 1000}{W_{solvent}(gm)} = \frac{n_{solute} \times 1000}{W_{solution} - W_{solute}}$$

$$= \frac{n_{solute} \times 1000 \times \frac{1}{V}\, solution\ (ml)}{\frac{W_{solution} - W_{solute}}{V_{solution}\ (ml)}}$$

$$m = \frac{M}{\frac{W_{solution}}{V_{solution}} - \frac{W_{solute}}{V_{solution}}}$$

$$m = \frac{M}{D - \frac{W_{solute}}{V_{solution}}} \quad \text{——— (1)}$$

$$M = \frac{W_{Solute}}{M \cdot M_{solute}} \times \frac{1000}{V_{solution(ml)}}$$

$$\frac{M(M \cdot M)_{Solute}}{1000} = \frac{W_{solute}}{V_{solution}}$$

from eqn. (1),

$$m = \frac{M}{D - \frac{M(M \cdot M)_{Solute}}{1000}}$$

$$m = \frac{1000M}{1000D - M(M \cdot M)_{Solute}}$$

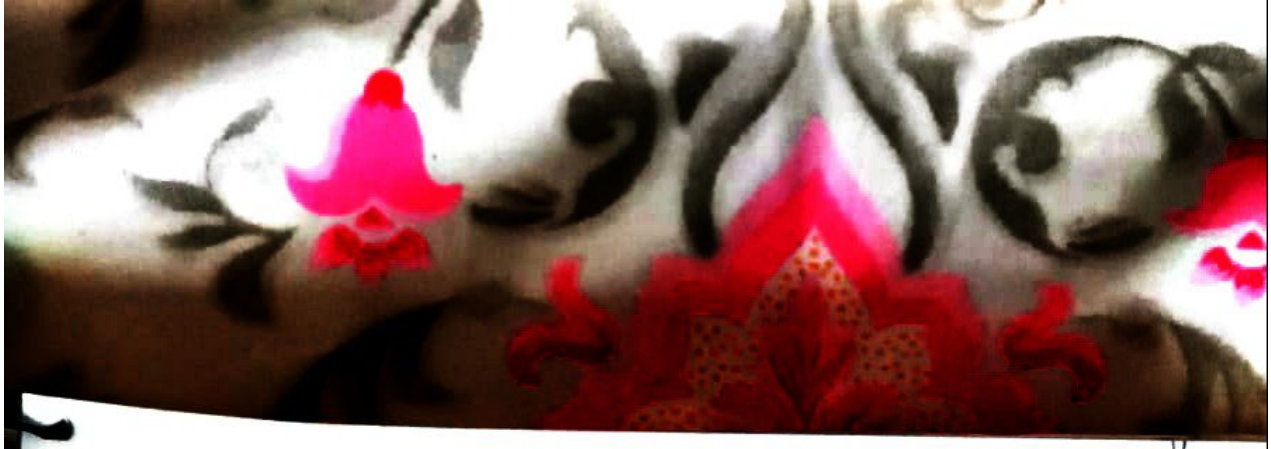
Note :- Molality is independent of temperature.

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(96) Pg - 46) Molals = 6.50, M.M = 56, D = 1.89

$$m = \frac{1000M}{1000D - M(M \cdot M Solute)}$$

$$6.50 = \frac{1000M}{1000 \times 1.89 - M(56)}$$



$6.50X(1890 - 56M) = 1000M$

$12285 - 364M = 1000M$

$12285 = 1364M$

$M = \frac{12285}{1364}$ 9.00 ...

$M = 9 \text{ mol dm}^{-3}$.

(rounded off to the nearest integer.)

Q) 16) pg-46)

$$m = \frac{1000M}{1000D - M(M \cdot M) \text{ Solute}}$$

$M = 0.8$

$D = 1.06$

$M \cdot M = 2 + 32 + 64$

$= 98$

$$m = \frac{1000 \times 0.8}{1000 \times 1.06 - 0.8 \times 98}$$

$$= \frac{800}{1060 - 78.4}$$

$$= \frac{800}{981.6} \; 0.8$$

$= 0.8$

$= 0.8 \times 10^{-3}$

$= 815$

* Illustrations :-

Pg-28 :-

Illustration 29:- 149 gm of potassium chloride (KCl) is dissolved in 10 Lt of an aqueous solution. Determine the molarity of KCl = 39 + 35.5 = 74.5 gm.

Sol) Moles of KCl = $\frac{\text{Mass (gms)}}{\text{Molar mass}}$

M.M of KCl = K ⇒ 39

Cl = 35.5

KCl = 74.5.

Moles = $\frac{149}{74.5}$ 2

= 2.

Molarity = $\frac{2}{10}$ = 0.2M

Illustration 30:- 225 gm of an aqueous solution contains 5gm of urea. What is the concentration of the solution in terms of molality?

Mol. wt of urea = 60.

Sol) Mass of urea = 5gm

Mol. wt = 60.

no. of moles in urea = $\frac{\text{mass (gms)}}{\text{mol. mass}}$

= $\frac{5}{60}$ × 0.083

= 0.083.

Mass of solvent = 225 - 5 = 220 gm.

Molality of solution = $\frac{\text{No of moles}_{\text{solute}}}{\text{Mass of}_{\text{solvent (gms)}}} \times 1000$

$= \frac{0.083}{\frac{220}{11}} \times \frac{50}{1000}$

$= \frac{4.150}{11}$

$= 0.332.$

Illustration 31 :— 0.5 g of a substance is dissolved in 25g of a solvent. Calculate the percentage amount of the substance in the solution.

Sol) Mass of substance = 0.5 g.
Mass of solvent = 25 g

$W/W\% = \frac{W_{solute\,(gm)}}{W_{(solution)\,(gm)}} \times 100$

$= \frac{0.5}{25+0.5} \times 100$

$= \frac{50}{25+0.5}$

$= \frac{50}{25.5}\ 10\ 1.96$

$= 1.96$

Illustration 32:- 20 cm^3 of an alchohol is dissolved in 80cm^3 of water. Calculate the percentage of alchohol solution.

Sol) Volume of alchohol = 20cm^3.
Volume of water = 80cm^3.

Percentage of alchohol solution = $\frac{20}{80+20} \times 100$

$= \frac{20}{100} \times 100$

$= 20\%$.

* Beginner's box - 6

Q2) What is molarity of H_2SO_4 solution that has a density of 1.84 g/cc and contains 98% by mass of H_2SO_4?

Sol) $M = \frac{(w/w\%)\, D\ gm/mL \times 10}{m \times M_{Solute}}$

$M = \frac{98 \times 1.84 \times 10}{98}$

$= 1.84 \times 10$

$= 18.4M$.

Q3) The molality of a sulphuric acid solution is 0.2. Calculate the total weight of the solution having 1000 gm of solvent.

Sol) $m = 0.2$.

$W_{solvent} = 1000$ gm

$$W_{solute} = \frac{0.2 \times 98 \times 1000}{1000}$$

$$= 19.6.$$

$W_{solution} = 19.6 + 1000$

$= 1019.6$ gms.

Q4) Calculate the molality of a sulphuric acid solution of specific gravity 1.2 containing 27% H_2SO_4 by weight.

Sol)

Density = 1.2 gm.

Weight of solution = 27%.

$H_2SO_4 = 98$.

$$\text{Molarity} = \frac{(w/w\%)\, D\, gm \times 10}{m \times m\,Solute}$$

$$= \frac{27 \times 1.2 \times 10}{98}$$

$$= \frac{27 \times 1.2 \times 5}{49}$$

$$= \frac{27 \times 6}{49}$$

$$= \frac{162}{49} = 3.3$$

$$\text{Molality} = \frac{1000 \times 3.3}{1000 \times 1.2 - 3.3 \times 98}$$

$$= \frac{3300}{1200 - 323.4}$$

$$= \frac{3300}{876.6} \quad \frac{1650}{550}$$

$$438.3$$

$$146.1$$

$$= \frac{550}{146.1} \quad 3.8$$

$$= 3.8$$

---

9) Calculate the molarity of each of the following:

a) 4g of caustic soda is dissolved in 200 ml of the solution.

$W_{solute} = 4g$

$W_{solvent} = 200\,ml$

$$m = \frac{4 \times 1000}{40 \times 200}$$

$$m = \frac{1}{2}$$

$$m = 0.5M$$

---

b) 5.3 g of anhydrous sodium carbonate is dissolved in 100 ml of solution.

$$m = \frac{5.3}{106} \times \frac{1000}{100}$$

$$m = 0.5M.$$

c) 0.365 g of pure HCl gas is solved in 50 ml of solution.

$$m = \frac{0.365}{36.5} \times \frac{1000}{50}$$

$$= \frac{1.46}{7.3}$$

$$= 0.2M$$

* Illustrations:-

Illustration 14:- Pg-34- 4 mole of $MgCO_3$ is reacted with 6 moles of HCl solution. Find the volume of CO_2 gas produced at the reaction is?

Sol) $MgCO_3 + \underset{L.R.A}{2HCl} \longrightarrow MgCl_2 + CO_2 + H_2O$.

No. of CO_2 moles produced = 3

Volume at STP = 3×22.4

= 67.2 lit.

Illustration 15:- Pg-34- 117 gm NaCl is dissolved in 500 ml aqueous solution. Find molarity.

Sol) $NaCl = 23 + 35.5 = 58.5$

$$m = \frac{117/58.5}{500/1000}$$

2

$$= \frac{117 \div 58.5}{2}$$

$$= \frac{117^2}{58.5} \times 2$$

$$= 2 \times 2$$

$$= 4M$$

Illustration 16:- 518 gm of an aquoer solution contains 18 gm of glucose (wt=180). Find molarity of the solution.

Sol) Wt of solvent = 518 - 18 = 500 gm.

$$\text{Molarity} = \frac{18/180 \times 10}{500/1000 \times 2}$$

$$= \frac{1}{10} \times 2$$

$$= 0.2M$$

Illustration 17:- 0.25 of a substance is dissolved in 6.25 gm of a solvent. what is % amount of the substance in the solution?

Sol) $Wt_{solution} = 0.25 + 6.25$

$= 6.5$

$$w/w\% = \frac{0.25}{6.5} \times 100$$

$$= \frac{25}{65} \times 3 \quad 3.8$$

$= 3.8\%$

* Mole fraction (x) :-

It is the ratio between no. of moles of one substance to all the substances present in the solution.

$$\underset{2}{\text{Solute}} + \underset{1}{\text{Solvent}} \longrightarrow \text{Solution}$$

$$X_2 = \frac{n_2}{n_1 + n_2} = \frac{n_2}{n_T}$$

where, $n_T = n_1 + n_2$

Mole fraction has no units (Dimensionless).

$$X_1 = \frac{n_1}{n_1 + n_2}$$

Note :- In a solution, sum of mole fraction of all the substances is equal to "1".

$$X_1 + X_2 = \frac{n_1}{n_1 + n_2} + \frac{n_2}{n_1 + n_2}$$

$$= \frac{n_1 + n_2}{n_1 + n_2}$$

$$= 1$$

$$X_1 + X_2 = 1$$

$$X_1 = 1 - X_2 \quad \text{(or)} \quad X_2 = 1 - X_1.$$

Ex :- Calculate mole number of 3.9 gms of benzene and 9.2 gms of Tolune present in a solution.

Given - Benzene - C_6H_6

Toluene - C_7H_8

$$X_B = \frac{n_B}{n_B + n_T}$$

$n_B = \frac{3.9}{78} = 0.05$ $\quad n_T = \frac{9.2}{92} = 0.15$

$X_B = \frac{0.05}{0.15} = \frac{1}{3} = 0.33$

$X_T = 1 - X_B = 1 - 0.33 = 0.66.$

* Relation between molality and mole fraction :-

$$m = \frac{n_{solute} \times 1000}{W_{solvent}(gms)}$$

$$m = \frac{n_{solute} \times 1000 \times \frac{1}{n_T}}{W_{solvent}/n_T}$$

$$m = \frac{(n_{solute}/n_T) \times 1000}{W_{solvent}/n_T}$$

$$= \frac{x_{solute} \times 1000}{\frac{n_{solvent} \cdot (Mol.mass)_{solvent}}{n_T}}$$

$$m = \frac{x_{solute} \times 1000}{x_{solvent} \cdot (Mol.Mass)_{Solv}}$$

$n_{solvent} = \frac{Mass(gm)}{M.M}$

Mass of solvent $= n_{solvent} \times mol.mass$

$$\boxed{m = \frac{x_{Solute} \times 1000}{(1 - x_{solute})(Mol.mass)_{solvent}}}$$

=) For aqueous solutions (when water is solvent),

$$\boxed{m = \frac{x_{Solute} \times 1000}{(1 - x_{solute}) \times 18}}$$

⇒ m, χ and w/w% are independent of T.
whereas, M, N, w/v%, v/v% are dependent on T.

⇒ Species having very less concentration but effective in measurement is expressed in terms of ppM and ppb.

⇒ PPM :- (parts per million)

$$PPM = \frac{\text{Weight of solute (gms)}}{\text{Weight of solution (gms)}} \times 10^6$$

(OR)

$$PPM = \frac{\text{Volume of solute (ml)}}{\text{Volume of solution (ml)}} \times 10^6$$

$W_{Solute} \ll$ weight of solvent.

hence, $W_{solvent} \approx W_{solution}$

$$PPM = \frac{W_{solute}\ (gms)}{W_{solvent}\ (gms)} \times 10^6$$

⇒ Relation b/n PPM and M :-

$$PPM = \frac{W_{solute}\ (gms)}{V_{solution}\ (ml)} \times 10^6$$

$$M = \frac{W_{solute}\ (gms)}{\text{mol. mass solute}} \times \frac{1000}{V_{solution}\ (ml)}$$

For water, D = 1 gm/ml.

$$M_{(mol\ mass)\ Solute} = \frac{W_{solute}\ (gms)}{V_{solution}\ (ml)} \times 10^3$$

$$ppM = M_{(mol\ mass)_{solute}} \times 10^3$$

* PPb :- Parts Per billion :-

$$PPb = \frac{W_{solute}\ (gms)}{W_{solution}\ (gms)} \times 10^9$$

Note:- For PPM and PPb,

$$W_{solvent} \cong W_{solution} \cong V_{solvent} \cong V_{solution}$$

Molarity = molality

* Eudiometry :-

It is a method of gas analysis formed in a reaction of at least two gas substances.

⇒ Eudiometry works on the principle of Lussac's law.

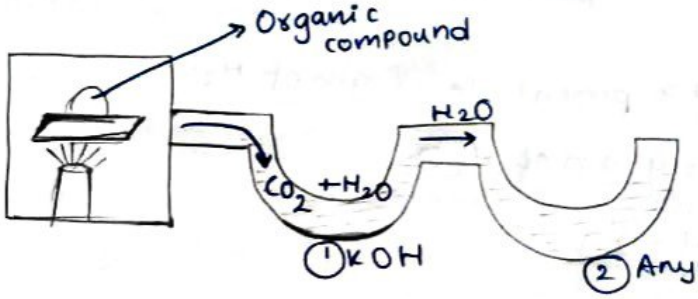
According to this law, in a gaseous reaction, the gases react in the simplest ration of their volume.

Ex: $H_{2(g)} + Cl_{2(g)} \rightarrow 2HCl$.

*	Absorbent	Gas.
1,	KOH	CO_2, SO_2, Cl_2
2,	$CuCl_2$	CO
3,	Turpentine oil	O_3
4,	Alkaline pyragallol	O_2

5.	Anyhydrous $CuSO_4$ (OR) Anyhydrous $CaCl_2$	H_2O
6.	Water	NH_3, HCl.

* % of Carbon and Hydrogen :-



After the complete combustion of organic compound the increase in mass of KOH contained is the amount of CO_2 formed and increase in mass of $CaCl_2$ contained is mass of H_2O formed.

$$C_xH_y + \left(x + \frac{y}{4}\right)O_{2(g)} \longrightarrow xCO_{2(g)} + \frac{y}{2}H_2O_{(g)}$$

* % of Carbon :-

$$\% \text{ of } C = \frac{\text{Mass of carbon (gm)}}{\text{Mass of org. compound}} \times 100$$

12 gm of carbon is present in 44gm of CO_2

Mass of carbon = x gm of CO_2

mass of organic compound = w gms

$$\text{Mass of carbon} = \frac{12x}{44}$$

% of carbon = $\frac{X}{W} \times \frac{12}{44} \times 100$

* % of H :-

$$\% \text{ of } H = \frac{\text{Mass of } H}{\text{Mass of org. compd}} \times 100$$

2 gm of H is present in 18 gm of H_2O.

Mass of H = y gm of H_2O

Mass of $H = \frac{2y}{18}$

$$\% \text{ of } H = \frac{Y}{W} \times \frac{2}{18} \times 100$$

Q) What volume of O_2 is needed for complete combustion of 40ml Ethane?

Sol) Ethane $\rightarrow C_2H_6$

1M Ethane →

$$C_2H_6 + \left(2 + \frac{6}{2}\right) O_{2(g)} \longrightarrow 2CO_2 + \frac{6}{2} H_2O_{(g)}$$

$$C_2H_6 + \frac{7}{2} O_2 \longrightarrow 2CO_2 + 3H_2O$$

1 ml of $C_2H_6 \cong 3.5$ ml of O_2

40 ml of $C_2H_6 \cong 40(3.5)$ ml of O_2

$\cong 140$ ml.

Q) Calculate the volume (in ml) of CO_2 evolved by the combustion of 50 ml of a mixture containing 40% C_2H_4 and 60% CH_4

Sol) 40% of 50 ml → 20 ml ethylene (C_2H_4)

$C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2H_2O$

20 ml C_2H_4 ≙ 40 ml of CO_2

30 ml of CH_4 ≙ 30 ml of CO_2

= 70 ml

Beginner's box - 6

Q1) Mole fraction of A in H_2O is 0.2. The molality of A in H_2O is :-

Sol) Mole fraction :- $\frac{X_{Solute} \times 1000}{(1 - X_{Solute}) \times 18}$

$$m = \frac{0.2 \times 1000}{0.8 \times 18}$$

$$m = \frac{2 \times 1000}{8 \times 18} = \frac{500}{250}$$

$$m = 13.9$$

Q6) A mixture of ethanol and water contains 54% by mass. Calculate the mole fraction of alcohol in this solution.

Sol) $M = \frac{(w/w\%) \times 1 \times 10}{18}$ 5 / 9

$= \frac{54 \times 5}{9} = 30M$

Ethanol → $C_2H_5OH = 24 + 5 + 16 + 1 = 46$

Converting from % to moles :-

$\Rightarrow \frac{54}{18} = 3$ moles of H_2O

$\Rightarrow \frac{46}{46} = 1$ mole of ethanol

$\Rightarrow$ mole fraction $= \frac{X_1}{X_1 + X_2}$

$= \frac{1}{1+3}$

$= \frac{1}{4} 0.25$

$= 0.25 X$

Q7) Mole fraction of ethanol in ethanol and water mixture is 0.25. Hence percentage concentration of ethanol by weight of mixture is :-

Sol)

Mole fraction = 0.25.

$0.25 = \frac{X_1}{X_1 + X_2}$

$X_1 = 0.25 X_1 + 0.25 X_2$

Ethanol = 0.25

Water = 0.75

Ethanol = C_2H_5OH = 46

Water = H_2O = 18

Mass percentage = $\frac{46}{100} \times 100$

= 46%

Q8) How many moles of Na^+ ions are in 20 ml of 0.40M of Na_3PO_4:-

Sol) No. of moles of Na_3PO_4 = 0.4 M

Molar mass of Na_3PO_4 = 16

$$\text{molarity} = \frac{\text{no. of moles}}{\text{vol(lt)}}$$

$$\text{no. of moles} = \text{molarity} \times \text{Volume (lt.)}$$

$$= \frac{0.4 \times 20}{1000} = \frac{0.1}{125} = \frac{1}{1250}$$

$$= 0.0080$$

3 moles of $Na^+ \cong$ 1 mole of Na_3PO_4

$$= 0.0080 \times 3$$

$$= 0.024 \text{ M.}$$

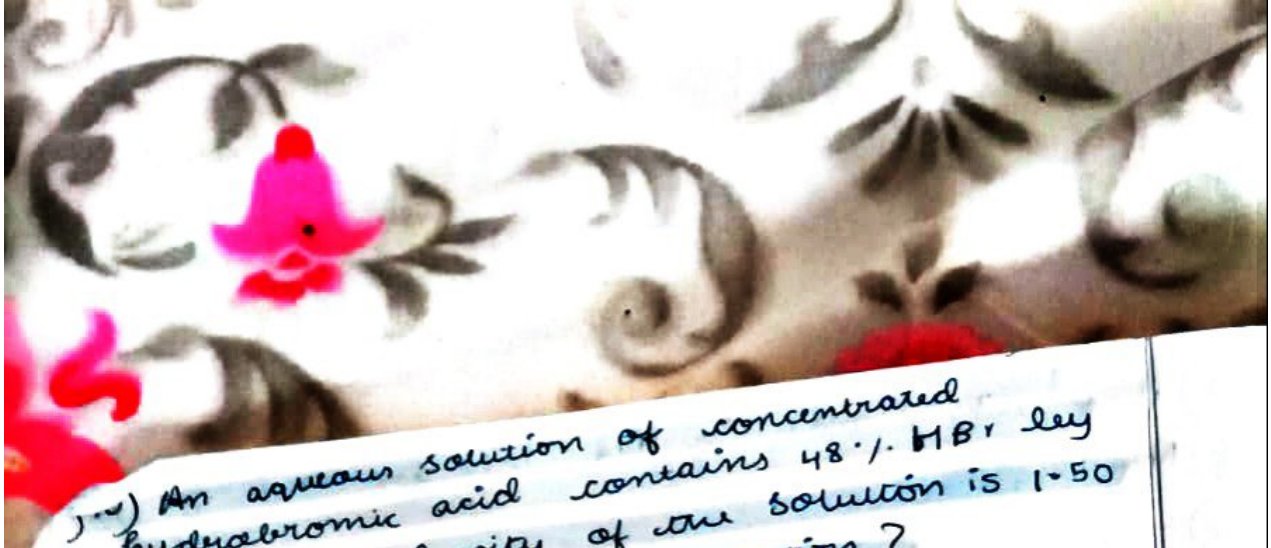
Q9) The molarity of the solution containing 2.8 % mass volume of KOH (Atomic mass of K = 39) :-

Sol) W/V % = 2.8 %

Molar mass = 39 + 16 + 1 = 56

$$M = \frac{(W/V) \times 10}{\text{m. mass}} = \frac{2.8 \times 10}{56} = \frac{28}{56} = 0.5$$

M = 0.5 M



Q11) An aqueous solution of concentrated hydrobromic acid contains 48% HBr by mass. If the density of the solution is 1.50 g/ml, what is its concentration?

Sol) mass of HBr = 48 %

density = 1.50 g/ml

$$\text{molarity} = \frac{(W/W\%)\, D \text{ gm/ml} \times 10}{m \times M \text{ solute}}$$

$$= \frac{48 \times 1.50 \times 10}{81}$$

$$= \frac{48 \times 15}{81}$$

$$= \frac{720}{81} = 8.9$$

$$= 8.9 \text{ mol/L}$$

Q12) Calculate the amount of pure 95% Na_2CO_3 required to prepare 5 litres of 0.5 M solution.

Sol)

$$M = \frac{n}{V}$$

$$n = m \times V$$

$$n = 0.5 \times 5$$

$$n = 2.5 \text{ mol}$$

molar mass $_{Na_2CO_3} = 106$

$$= \frac{95}{106} = 106 \times 2.5$$

$$= 265.0$$



Carbonate = $\frac{53}{19} \times 100$

$= \frac{53}{19} \times 100$

$= 2.789 \times 100$

$= 278.9$ gms